

4.5
1 $h(k) = k \bmod 97$

a) $034567981 \bmod 97 = \boxed{91}$

b) $183211232 \bmod 97 = \boxed{57}$

c) $220195744 \bmod 97 = \boxed{21}$

d) $987255335 \bmod 97 = \boxed{5}$

4.5
3a $h(k) = k \bmod 31$

$317 \bmod 31 = \boxed{7}$

$918 \bmod 31 = \boxed{19}$

$007 \bmod 31 = \boxed{7}$

$100 \bmod 31 = \boxed{7}$

$111 \bmod 31 = \boxed{18}$

$310 \bmod 31 = \boxed{0}$

4.5
19b $8+8+3+8+2+0+1+3+4+4 = 41$

$41 \bmod 9 = 5 \checkmark$

4.5
20b $6+7+0+2+1+2+0+Q+9+8 \bmod 9 = 8$

$35 + Q \bmod 9 = 8$

$35 \bmod 9 + Q \bmod 9 = 8$

$8 + Q \bmod 9 = 8$

$Q \bmod 9 = 0$

$Q = 0 \text{ or } Q = 9$

can't tell

4.5
21b

$$8 + 5 + 0 + Q + 9 + 1 + 0 + 3 + 8 + 5 \pmod{9} = 8$$

$$39 + Q \pmod{9} = 8$$

$$39 \pmod{9} + Q \pmod{9} = 8$$

$$3 + Q \pmod{9} = 8$$

$$Q \pmod{9} = 5$$

$$\boxed{Q = 5}$$

4.5
24b

$$3 \cdot 6 + 3 + 3 \cdot 6 + 2 + 3 \cdot 3 + 9 + 3 \cdot 9 + 1 + 3 \cdot 3 + 4 + 3 \cdot 6 + X_{12} = 0 \pmod{10}$$

$$18 + 3 + 18 + 2 + 9 + 9 + 27 + 1 + 9 + 4 + 18 + X_{12} = 0 \pmod{10}$$

$$118 + X_{12} = 0 \pmod{10}$$

$$\boxed{X_{12} = 2}$$

4.5
28a

$$10237424413392 \pmod{7} = \boxed{1}$$

4.5
33

a) $3 \cdot 1 + 4 \cdot 0 + 5 \cdot 5 + 6 \cdot 9 + 7 \cdot 1 + 8 \cdot 0 + 9 \cdot 2 \pmod{11} = 107 \pmod{11} = 8 \quad X$

b) $3 \cdot 0 + 4 \cdot 0 + 5 \cdot 0 + 6 \cdot 2 + 7 \cdot 9 + 8 \cdot 8 + 9 \cdot 9 \pmod{11} = 220 \pmod{11} = 0 \quad \checkmark$

c) $3 \cdot 1 + 4 \cdot 5 + 5 \cdot 3 + 6 \cdot 0 + 7 \cdot 8 + 8 \cdot 6 + 9 \cdot 6 \pmod{11} = 196 \pmod{11} = 9 \quad \checkmark$

d) $3 \cdot 1 + 4 \cdot 0 + 5 \cdot 0 + 6 \cdot 7 + 7 \cdot 1 + 8 \cdot 2 + 9 \cdot 0 \pmod{11} = 68 \pmod{11} = 2 \quad X$

ABCDEFGHIJKLMNOPQRSTUVWXYZ

4.6
1 DO NOT PASS GO

a) $f(p) = (p+3) \bmod 26$ GR QRV RDVV JR

b) $f(p) = (p+13) \bmod 26$ QB ABC CNFF TB

c) $f(p) = (3p+7) \bmod 26$ QX UXM AHJJ ZX

4.6
3 WATCH YOUR STEP

a) $f(p) = (p+14) \bmod 26$ KOHQV MCIF GHSD

b) $f(p) = (14p+21) \bmod 26$ RVBXP TJPZ NBZX

c) $f(p) = (-7p+1) \bmod 26$ DBYNE PHRM FYZA

4.6
4 a) EOXH MHDQV
BLUE JEANS

b) WHVW WRGDB
TEST TODAY

c) HDW GLP VXP
EAT DIM SUM

4.6
6 $f(p) = (p+k) \bmod 26$

"E" is the most common letter in English, so we choose $k = 19$
to map E to X.

4.6
8 MEN LOVE TO WONDER, AND THAT IS THE SEED
OF SCIENCE.

JVVU $\rightarrow$ V is probably a vowel, probably E or O.
KXV, JTZVETV $\rightarrow$ V at the end of words. E is
more likely than O.

Thus, try $V \rightarrow E$ $f(p) = (p+17) \bmod 26$

4.6
14 GRIZZLY BEARS

GRIZZ LYBEA RSXXX

IZGZR BELAY XXRXS

4.6
15 EABW EFRO ATMR ASIN

BEWA REUF MART IANS

BEWARE OF MARTIANS

4.6
17 Shift Cypher

9.1
1a

$(0,0) (1,1) (2,2) (3,3)$

9.1
3a

$\{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$

$S = \{1, 2, 3, 4\}$

not reflexive, missing $(1,1)$

not symmetric, missing $(4,2), (4,3)$

not antisymmetric, have $(2,3), (3,2)$

transitive: $(2,3), (3,2) \rightarrow (2,2)$ $(3,2), (2,3) \rightarrow (3,3)$

9.1
5

a) reflexive: everyone who has visited a has also visited a

not symmetric: could be the case that the people who visited a are a proper subset of people who visited b

not antisymmetric: could be that page a and b have exactly the same set of visitors

transitive: if everyone who has visited a has visited b, and everyone who visited b has visited c, then everyone who has visited a has visited c.

b) not reflexive: a has the same links as a

symmetric: a and b share no links $\equiv$ b and a share no links

not antisymmetric: there probably exists two pages a and b that share no links

not transitive: if a and b share no links and b and c share no links, a and c could still share links.

c) not reflexive: a page could have 0 links on it.

symmetric: a and b have a common link $\equiv$ b and a have a common link

not antisymmetric: there probably exists two pages a and b that have a common link

not transitive: if a and b have a common link and b and c have a common link, a and c could share no links in common.

d) not reflexive: there are probably web pages with no links to them

symmetric: a page with links to a and b $\equiv$ a page with links to b and a.

not antisymmetric: there probably exists two pages a and b that are linked together on a page c.

not transitive: if a page links a and b and another page links b and c, there may not be a page that links a and c.

9.1
6c) $x-y$ is a rational number

reflexive: $x-x=0 \in \mathbb{Q}$

symmetric: if $x-y=r \in \mathbb{Q}$, then $y-x=-(x-y)=-r \in \mathbb{Q}$

not antisymmetric: $(1,-1) \in R$ and $(-1,1) \in R$

transitive: if $x-y=r_1 \in \mathbb{Q}$ and $y-z=r_2 \in \mathbb{Q}$, then

$$x-z=(x-y)+(y-z)=r_1+r_2 \in \mathbb{Q}.$$

9.1
7

a) $x \neq y$

not reflexive: $x \neq x$ is false

symmetric: $x \neq y \equiv y \neq x$

not antisymmetric: $1 \neq 0$

not transitive: $1 \neq 2$ and $2 \neq 1$, but $1=1$.

b) $xy \geq 1$

not reflexive: $0 \cdot 0 \not\geq 1$

symmetric: $xy \geq 1 \equiv yx \geq 1$

not antisymmetric: $2 \cdot 1 \geq 1$ and $1 \cdot 2 \geq 1$

transitive: if $ab \geq 1$ and $bc \geq 1$, then

case 1: a is positive $\rightarrow b$ is positive $\rightarrow c$ is positive,
so $ac \geq 1$

case 2: a is negative $\rightarrow b$ is negative $\rightarrow c$ is negative,
so $ac \geq 1$

c) $x = y + 1$ or $x = y - 1$

not reflexive: $1 \neq 1 + 1$ and $1 \neq 1 - 1$

symmetric: if $x = y + 1$, then $y = x - 1$ and if $x = y - 1$, then $y = x + 1$.

not antisymmetric: $2 = 1 + 1$ and $1 = 2 - 1$.

not transitive: $2 = 1 + 1$ and $1 = 2 - 1$, but $1 \neq 1 + 1$ and $1 \neq 1 - 1$.

d) $x \equiv y \pmod{7}$

reflexive: $x \equiv x \pmod{7}$

symmetric: if $x \equiv y \pmod{7}$ then $y \equiv x \pmod{7}$.

not antisymmetric: $2 \equiv 9 \pmod{7}$ and $9 \equiv 2 \pmod{7}$

transitive: if $x \equiv y \pmod{7}$ and $y \equiv z \pmod{7}$, then $x \equiv z \pmod{7}$.

e) x is a multiple of y

reflexive: $x = 1 \cdot x$

not symmetric: 6 is a multiple of 2, but 2 is not a multiple of 6

not antisymmetric: $2 = (-1)(-2)$, $(-2) = (-1)(2)$, but $-2 \neq 2$.

transitive: if $x = my$ and $y = nz$, then $x = mnz$.

f) x and y are both negative or nonnegative.

reflexive: x must be the same as itself.

symmetric: x and y negative/nonnegative $\equiv y$ and x negative/nonnegative

not antisymmetric: -5 and -6 both negative and $-5 \neq -6$

transitive: if x and y both negative/nonnegative, and y and z are both negative/nonnegative, then x and z must both be negative/nonnegative.

g) $x = y^2$

not reflexive: $2 \neq 2^2$

not symmetric: $4 = 2^2$ but $2 \neq 4^2$

antisymmetric: $x = y^2$ is only true when $1 = 1^2$ and $0 = 0^2$ which means $x = y^2$ only when $x = y$.

not transitive: $16 = 4^2$ and $4 = 2^2$ but $16 \neq 2^2$.

h) $x \geq y^2$

not reflexive: $2 \not\geq 2^2$

not symmetric: $2 \geq 1^2$ but $1 \not\geq 2^2$

antisymmetric: the only time we have $x \geq y^2$ and $y \geq x^2$ is when $x = y = 1$ or $x = y = 0$, meaning when $x = y$.

transitive: if $x \geq y^2$ and $y \geq z^2$, then

$$x \geq y^2 \Rightarrow x \text{ is nonnegative}$$

$$y \geq z^2 \Rightarrow y \text{ is nonnegative}$$

$$x \geq z^4 \geq z^2 \Rightarrow x \geq z^2$$

9.1
8

not reflexive: $\forall a \in S, (a, a) \notin R$

symmetric if $(a, b) \in R$, then $(b, a) \in R$

False, since $R = \emptyset$.

$\Rightarrow$ vacuously true

transitive if $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$

False, since $R = \emptyset$

$\Rightarrow$ vacuously true

9.1
14

a) no, $0 + 0 = 0$

e) no, $0 \cdot 0 \geq 0$

b) no, $0 = \pm 0$

f) no, 0 and 0 both nonnegative

c) no, $0 - 0 = 0 \in \mathbb{Q}$

g) no, $0 = 0^2$

d) no, $0 = 2 \cdot 0$

h) no, $0 \geq 0^2$

9.1
21

a) no, $1 + (-1) = 0$ and $-1 + 1 = 0$

b) no, $1 = -(-1)$ and $-1 = -1$

c) no, $1 - (-1) \in \mathbb{Q}$ and $-1 - 1 \in \mathbb{Q}$

d) no, $0 = 2 \cdot 0$ and $0 = 2 \cdot 0$

e) no, $2 \cdot 1 \geq 0$ and $1 \cdot 2 \geq 0$

f) no, $0 \cdot 1 = 0$ and $1 \cdot 0 = 0$

g) no, $1 = 1$ and $1 = 1$

h) no, $1 = 1$ and $1 = 1$

9.1
27

$$R = \{ (a, b) \mid a \text{ divides } b \} \quad \text{on } \mathbb{Z}^+$$

a) $R^{-1} = \{ (a, b) \mid b \text{ divides } a \}$

b) $\overline{R} = \{ (a, b) \mid a \text{ does not divide } b \}$